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Devices for integrated indirect sweat stimulation and sensing

Abstract

A sweat sensing device (20) comprises at least one sweat generation unit (22) capable of initiating sudomotor axon reflex (SAR) sweating in an indirect stimulation region and at least one analysis unit (24, 26) capable of sensing a physiological parameter of sweat, collecting a sweat sample, or a combination thereof. The at least one analysis unit (24, 26) is located above the indirect stimulation region when the sweat sensing device is placed on skin.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to and the benefit of the filing date of U.S. Patent Application Publication No. 2018/0035928 A1 (U.S. patent application Ser. No. 15/544,725), filed on Jul. 19, 2017, which is the U.S. national phase of International Patent Application No. PCT/US2016/017726 (International Patent Application Publication No. WO 2016/130905 A1), filed on Feb. 12, 2016, which claims benefit of U.S. Patent Application No. 62/115,851, filed on Feb. 13, 2015, the disclosures of which are hereby incorporated by reference herein in their entireties.

BACKGROUND OF THE INVENTION

(1) Historically, partitioning of biomarkers from blood to sweat has been demonstrated in great detail. As more of these biomarkers emerge, sweat appears to be a convincing media for continuous and spontaneous health monitoring. However, ‘clinical’ techniques of sample collection, purification and analysis have restricted growth in the sweat-sensing area because of the cost and time associated with these techniques. With the advent of miniaturized sensors, however, many of these issues can be alleviated. Still, a large task has largely been left unexplored for compact sweat sensing technologies: sample extraction and collection.

(2) Common techniques for sweat stimulation and analysis involve sweat stimulation in a region from a sweat generating unit **10**, followed by removal of this unit **10** from the skin **12**, cleaning of the skin **12** and reapplication of a sensing unit **14** or collection device **16**, as shown in FIGS. **1A-1C**. Often, the sensing unit **14** has integrated communication protocol to alert the

user. This communication method could be via wireless or wired connections. For example, cystic fibrosis testing often involves this technique as demonstrated by ELITechGroup® in their Macroduct® and Nanoduct® products. This technique is problematic because it requires a two-step process that is inconvenient, non-continuous, and where reproducibility could be difficult.

(3) Further, contamination from the stimulation reservoir and sensor region is unavoidable as they share the same area. In reference to FIGS. 1A-1C, the sweat generation unit **10** has potential to alter the state of the skin **12**, whether that includes excessive hydration, thermal heating, irritation, pharmacological side-effects, or another side effect. This adds a confounding factor to sensing sweat analytes when the sensing unit **14** or collection device **16** is placed on the skin.

(4) Attempts to reduce contamination have previously been made. For example, a technique includes utilizing an isolating membrane between sweat stimulation mechanisms and the sensors and sensing sites. However, such techniques utilizing isolating membranes (or similar techniques) may only partially or temporarily separate sweat stimulation mechanisms, such as an electric field and/or chemicals, from the sensors and sensing sites. In the instance of isolating membranes, these also have the drawback of increasing the dead volume between a sensor and the skin **12**, which reduces temporal resolution. Furthermore, horizontal iontophoretic driving of an iontophoretic chemical, such as pilocarpine, may be used. However, this will again subject the sensor, sweat, and skin to an electric field and/or contamination. For many biomarkers and sensors, such interference could reduce performance of a sweat sensing device, in some cases making sensing impossible. There is an increasing need to provide improved sweat sensing techniques and devices that address one or more of the above drawbacks.

SUMMARY OF THE INVENTION

(5) Embodiments of the present invention rely in part on the premise that sudomotor axon reflex (SAR) sweating can be utilized by a sweat inducing and sensing device for sweat analysis. SAR sweating can potentially be initiated by a variety of mechanisms: thermal, direct-electrical, chemical, occlusion, and others. In this setting, direct-electrical refers to a biophysical phenomenon where sweating is initiated by the flow of electron and ion current without the aid of a chemical active compound. Additionally, embodiments of the present invention can greatly reduce contamination and improve chronological sampling of sweat. Furthermore, the embodiments of the present invention described below have the ability to use a variety of sensing techniques which greatly improves the impact and variety of applications for such a device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The objects and advantages of the present invention will be further appreciated in light of the following detailed descriptions and drawings in which:

(2) FIGS. 1A-1C are cross-sectional views of a prior art sweat stimulation, sensing, and collection technique;

(3) FIG. 2 is a cross-sectional view of a device according to an embodiment of the present invention showing separate sensing and collection units;

(4) FIG. 3 is a cross-sectional view of a device according to another embodiment of the present invention showing the sensing and collection units embedded within the sweat generation unit;

(5) FIG. 4A is a top view of a device according to another embodiment of the present invention showing a hexagonal arrangement;

(6) FIG. 4B is a cross-sectional view of the device of FIG. 4A;

(7) FIG. 5 is a top view of a device according to another embodiment of the present invention showing a high aspect ratio arrangement;

(8) FIGS. 6A and 6B are cross-sectional views of devices according to embodiments of the present invention showing a fluid management system in differing arrangements; and

(9) FIGS. 6C and 6D are perspective views of devices according to embodiments of the present invention showing a fluid management system in differing arrangements.

DETAILED DESCRIPTION OF THE INVENTION

(10) The embodiments of the present invention described herein improve greatly on prior simultaneous stimulation and collection of sweat sample methods, such as that presented in FIG. 1.

(11) Embodiments of the present invention take advantage of a biological response referred to as the sudomotor axon reflex (“SAR”). This mechanism acts on the premise that innervation of sweat glands occurs as a result of peripheral functionality of sudomotor units (i.e., the body will stimulate a series of sweat glands directly underneath the stimulation region, “direct stimulation”, but will also generate sweat from sweat glands outside of this region, “indirect stimulation”). For example, in the case of chemical sweat stimulation, the sweat stimulant acts on the neural receptors surrounding the sweat glands to elicit a sweating response. The chemical stimulant can act on two primary receptors at the base of the sweat gland: muscarinic or nicotinic receptors. SAR response is typically observed with chemicals that interact strongly with nicotinic receptors at the base of the sweat gland. For example, pilocarpine acts weakly on nicotinic receptors and is therefore a poor chemical stimulant for SAR response. However, nicotine acts strongly on nicotinic receptors and is therefore an attractive stimulant for SAR response. Furthermore, there are chemical stimulants, such as acetylcholine, that act strongly on both muscarinic and nicotinic receptors, which can be leveraged to produce a SAR response. It should be recognized that, although not named, multiple other chemical stimulants are capable of causing a SAR response and are useful in embodiments of the present invention. Although there has largely been limited research in this area of sweat stimulation, it is also hypothesized that thermal, direct-electrical, occlusion, and other sweat stimulant techniques will produce a similar antidromic responses as

chemical stimulants. Consequently, one can stimulate sweat glands in a region within close proximity of a sensor array to generate sweat directly underneath the stimulation region (“direct stimulation”) and directly underneath the sensors (“indirect stimulation”). Typically, the “spreading” of SAR induced sweating, the distance from the edge of direct stimulation to the decay of indirect stimulation, is limited on the order of several tens of millimeters (e.g., up to about 30 mm). The degree of a SAR response, in the case of chemical stimulation, depends largely on the amount and type of sweat stimulant which is delivered to a given location.

(12) In reference to FIG. 2, a device **20** according to an embodiment of the present invention may have a structure with a sweat generation unit **22** and at least one of two analysis units: a collection unit **24** and a sensing unit **26**. The sweat generation unit **22** is capable of directly initiating sweat in a direct stimulation region and initiating SAR sweating in indirect stimulation regions under the collection unit **24** and the sensing unit **26**. In one embodiment, the sweat generation unit **22** includes a chemical stimulant, such as acetylcholine, methacholine, nicotine, carbachol, or another chemical that is capable of initiating SAR sweat. The collection unit **24** and the sensing unit **26** are located separately from the sweat generation unit **22** along the skin **12**. In one embodiment, the analysis units may be spaced apart from the direct stimulation region by a distance of about 0.1 mm to about 30 mm. The SAR-initiated sweat is then collected and sensed by the collection unit **24** and the sensing unit **26**, respectively. As shown, embodiments of the present invention promote a single-step process, which is an advantage over previous devices where sweat generation and sweat analysis were performed in multiple steps. Further, continuous and multiple occurrence SAR-initiated sweating for the purpose of sweat analysis is possible. For example, sweat may be sensed or collected in one instance in time, continuously for a length of time, or a combination thereof.

(13) Optionally, each of these regions (sweat generation, sweat sensing, and sweat collecting) may be separated by an isolation layer **28** as shown in FIG. 2. This isolation layer **28** could take many forms or materials such as an adhesive or rubber with the purpose of electrically and/or fluidically isolating regions of the device. The isolation layer **28** serves to improve the reliability of the device and helps assure that fluid will not mix between regions or create unwanted electrical connections (e.g., “electrical shorts”). Furthermore, since this isolation layer **28** is preferably water-insoluble, common adhesives will not only provide electric and fluidic isolation but will also aid in keeping the device on the skin **12**. Furthermore, in some cases, the isolation layer **28** protects the sensing unit **26** from voltages or currents applied to the sweat generation unit **22**.

(14) The term “sweat generation unit”, including other denotative or connotative phrases, as used herein, captures a plurality of sweat stimulation methods that are capable of initiating SAR sweating. For example, a sweat generation unit may involve one or more of chemical, thermal, direct-electrical, or other suitable mechanisms which stimulates the generation of sweat and are not specifically described. The most common technique for sweat stimulation is a chemical technique referred to as iontophoresis. This involves electric-field driven movement of a sweat stimulant drug into the skin surface, ultimately reaching the secretory coil of the sweat gland, to initiate sweating. It should be recognized that, although iontophoresis is the most common chemical technique, electroporation, injection or microneedles delivery, passive diffusion of a sweat stimulant from a drug reservoir, which may be improved by a diffusion enhancer (e.g., propylene glycol) applied to the skin prior to device application or incorporated directly into the stimulation unit itself, or other techniques are also possible routes of chemical delivery of a sweat stimulant in embodiments of the present invention. Utilizing such a design according to the present invention will greatly reduce contamination between a stimulation reservoir (e.g., in the sweat generation unit **22**) and collection and/or the sensor region (e.g., relating to the collection unit **24** and the sensing unit **26**).

(15) Additionally, the terms “sensing unit” and “sensing mechanism,” including other denotative or connotative phrases, as used herein could include one or more of a plurality of mechanisms for sensing sweat and/or its components or properties including potentiometric, amperometric, conductometric, impedance spectroscopy, skin impedance, galvanic skin response (GSR), or other suitable mechanisms. Similarly, the term “collection unit”, including other denotative or connotative phrases, as used herein, describes a collection method, material, or structure.

(16) The terms “collection and/or sensing unit” and “analysis unit”, including other denotative or connotative phrases, as used herein, describe a unit that is capable of sensing sweat, collecting sweat, or a combination of the two. A sensing unit (e.g., sensing unit **26**) and/or a collection unit (e.g., collection unit **24**) may have integrated electronics or controls which monitor physiological parameters, provide feedback to a user or similar function. In an embodiment with both a sensing unit(s) and a collection unit(s) (e.g., device **20**), the units may function independently of each other or may operate together.

(17) Sweat generation units, collection units, sensing units, and combinations of such units may include a variety of functional aspects such as wired or wireless communications, rigid or flexible structure or other method, material, function, or particular structure not specifically described here. These units may also have intelligent communication between or within each unit via optical, electrical, or similar communication method (not shown).

(18) In another embodiment, FIG. 3 shows a device **30** where a sweat generation unit **32** is placed in vertical alignment with a collection unit **34** and a sensing unit **36**. Although the collection **34** and/or sensing unit **36** are shown being located within the sweat generation unit **32**, another embodiment may include at least one collection **34** and/or sensing unit **36** located directly on skin **12** with the sweat generation unit **32** placed directly over said components (not shown). Although a minimal amount of direct sweating may possibly be produced underneath the collection unit **34** or sensing unit **36** from the sweat generation unit **32**, indirect (SAR) sweating may be the only suitable mechanism for generating sufficient sweat to sense and/or collect. For example, in the case of chemical stimulation via iontophoresis, the collection unit **34** and the sensing unit **36** reduce the effectiveness for direct stimulation directly underneath these units. Essentially, these units block or alter the effectiveness of a stimulation technique that initiates non-SAR sweating. However, in leveraging SAR stimulation, the device **30** will overcome these limitations by indirectly generating sweat underneath the units **34**, **36**. In other words, to generate sufficient sweat under the collection unit **34** and the sensing unit **36**, the sweat generation unit **32** includes a mechanism for

initiating SAR sweating.

(19) The construction of device **30** could simplify device construction and assembly compared to other configurations. For example, in one embodiment, units **34**, **36** could be fabricated using standard flexible electronics techniques (such as on PET or Kapton film), and pressed against skin **12**. The sweat generation unit **32** could be a gel including a sweat stimulant and a driving electrode (not shown) that is pressed down against units **34**, **36** and skin **12**. Some of the iontophoretic chemical stimulant in sweat generation unit **32** may find itself between units **34** and **36** and skin (similar to as diagramed in FIG. 3), but would be incapable of generating a strong direct stimulation of sweat, and thus this example embodiment would rely on the indirect stimulation of sweat by SAR to cause sweat to be received by units **34**, **36**.

(20) A benefit of the vertical alignment between the analysis units (e.g. collection unit **34** and sensing unit **36**) and a sweat generation unit (e.g., sweat generation unit **32**) is an increase in the density of the units. This benefit could be realized in other configurations of the sweat generation and analysis units, such as in the honeycomb formation described below. Depending on the sweat generation mechanism, SAR-initiated sweat may only be able to be collected or sensed up to several millimeters away from the direct stimulation region. Therefore, embodiments of the present invention achieve a greater benefit with a high density of sweat generation and analysis units.

(21) Regarding FIGS. **4A** and **4B**, in one embodiment, a device **40** is shown positioned on the skin **12**. The device **40** includes a plurality of sweat generation units **42** and a plurality of collection and/or sensing units **44** arranged in a honeycomb structure. In one embodiment, the sweat generation units **42** are iontophoretic in nature and the collection and/or sensing units **44** include potentiometric sensors. In this configuration, the collection and/or sensing units **44** are placed spatially so that each unit **44** is within close proximity of a sweat generation unit **42** (i.e., the units **44** are located above indirect stimulation regions). With the hexagonal arrangement of device **40**, each sensing unit **44** is surrounded by three sweat generation units **42**, increasing the probability for a SAR sweating response underneath the sensing units **44**. The device **40** further includes an optional isolation material **46** (best shown in FIG. **4B**) that isolates the collection and/or sensing units **44** from the sweat generation units **42**, which improves the device integrity and functionality. This isolation material **46** could take one of many forms and materials as previously described. Those of ordinary skill in the art will recognize that the structure may be configured to be a similar shape other than a hexagonal or a honeycomb structure. As shown in FIG. **4A**, the stimulation source (i.e., the sweat generation units **42**) and the collection and/or sensing units **44** are separate in horizontal location on skin, with no horizontal overlap. The present invention also contemplates devices where at least partial overlap exists where the benefit of SAR sweating in the present invention is still realized, such as device **30** presented in FIG. 3.

(22) Furthermore, in a device according to the present invention where the sweat generation method is via direct-electrical methods or iontophoresis, a return electrode can be placed on the periphery of the device so as to maximize the amount of current or drug delivered in the desired location. For example, in FIG. **4A**, a return electrode **48** is positioned around an edge of the device **40**.

(23) In FIG. 5, a portion of a device **50** according to an embodiment of the present invention is shown. The device **50** includes a single large area or a plurality of sweat generation units **52**, a plurality of collection and/or sensing units **54**, and an optional isolation material **56**. The structure utilizes a greater than unity aspect ratio of sensors and/or collection units **54**, which minimizes the amount of lateral distance required to leverage SAR sweating. Due to the need to be above at least one active sweat gland, and/or due to signal to noise requirements, some collection and or sensing units must have a minimum total area of contact with skin. A greater than unity aspect ratio allows a minimum area to be achieved, while further minimizing the amount of lateral distance required to leverage SAR sweating. In one embodiment, the length/width aspect ratio may be about 2:1 or, alternatively, about 1:2. Although FIG. 5 shows a rectangular configuration, those of ordinary skill in the art will recognize that a rectangular configuration is not required and a similarly intended effect can be achieved via a concentric design or similar structure (not shown). For example, other geometries, such as a star shape, may also be included that feature a greater than unity aspect ratio. Where the sweat generation method for the device **50** is via direct-electrical methods or iontophoresis, a return electrode **58** can be placed on the periphery of the device **50** so as to maximize the amount of current or drug delivered in the desired location.

(24) With reference to FIGS. **6A-6D**, devices according to various embodiments are shown that include improved sweat fluid management. More particularly, devices **60a**, **60b**, **60c**, and **60d** are shown including sweat generation unit(s) **62**, collection and/or sensing unit(s) **64**, and isolation materials **66**. An optional wicking material **68** is used as an additional component that acts as a fluid management feature, which improves performance. The wicking material **68** may also be used to bring sweat from skin **12** to a collection and/or sensing unit **64**, as described below. A device including the wicking material **68** may have varying configurations. In one embodiment, the wicking material **68** may act to actively move previously analyzed sweat sample to a disposal region (not shown) thereby increasing effective sweat sampling time resolution. For example, the wicking material **68** may be included as a through-hole component in the collection and/or sensing unit **64** (FIG. **6A**), at the edge of the collection and/or sensing unit **64** (FIG. **6B**), or in the plane of the collection and/or sensing unit **64** (FIG. **6C**). Further, this wicking material **68** or similar microfluidic structure could be used to transport sweat from the skin to the collection and/or sensing unit(s) **64** (specifically shown in FIG. **6D**). For ease of fabrication or intended application, one particular structure may be more beneficial than another. For instance, regarding device **60a**, there may be benefit in wicking sweat to a central location within the collection and/or sensing unit **64** to reduce likelihood of fluid buildup near the isolation boundary **66**. Similarly, in device **60c**, the wicking material **68** is shown partially extended along the length of the sensor and/or collection unit **64**, which may prove beneficial where a small volume of sweat is generated underneath the sensor and/or collection unit **64** before being wicked away. In this instance, the wicking material **68** acts as a static, volume-limiting pump. Furthermore, a combination of one of the structures described may prove beneficial. In an exemplary embodiment, the wicking material **68** may be paper or a polymer or microfluidic feature that operates via capillary action. Further, the wicking

material **68** could be segmented utilizing more than one wicking material, such as a combination of paper and polymer or another combination of similar wicking materials.

(25) Prediction of Typical Parameters

(26) Peripheral (indirect) sweating after 5 minutes of iontophoresis is significant (e.g., sweat rates of about 1-3 nL/min/gl) on average to at least 8 mm from the boundary of the direct stimulation region. A sweat stimulant could be delivered iontophoretically or by another method and could constitute a wide range of various sweat stimulants (e.g., pilocarpine, acetylcholine, etc.). Considering that sensors can be fabricated utilizing previously demonstrated technology well below 8 mm in diameter (e.g., 500 μm in diameter), an array of biomarker sensors may be easily placed within the indirect sweating region. Although indirect sweating results in a reduction in sweat rate (e.g., about 60% reduction as compared to the sweat rate in the area being directly stimulated), the reduction of the risk of contamination and potential to reduce drug dosage (as described below) outweigh this downfall.

(27) The spacing between the stimulation regions and the collection and/or sensor units may vary depending on the configuration. In one embodiment, an array of stimulation units and analysis units may have a minimum half-pitch (smallest repeating unit) less than one millimeter at an accuracy within tens of micrometers or better.

(28) Sweat Stimulation—Sampling Effectiveness

(29) A sampling effectiveness metric (“SE”) for comparing sweat stimulation devices can be defined by Equation 1:

$$(30) SE = \frac{Q}{\Gamma}$$

where Q is the sweat rate per minute, and Γ is the dose of drug stimulant delivered in the case of iontophoretic delivery. A base metric, e.g., SE.sub.0, for the case of the stimulation region being the same as the collection region (e.g., as shown in FIGS. 1A-1C) may be used to compare sweat stimulation devices without using actual sweat rate data. The hexagonal device **40** shown in FIGS. 4A and 4B have one stimulation unit **42** per four total stimulation and collection/sensor units **42**, **44** per unit cell and would utilize at least $\frac{1}{4}$, or 25%, of the stimulation region and, thus, 25% of the dosage $\Gamma_{\text{sub.0}}$. Similarly, the sensing/collection regions would comprise 75% of the original area, $A_{\text{sub.0}}$. Further, as described above, the hexagonal device **40** would generate 40% of the original sweat rate, $Q_{\text{sub.0}}$, in the indirect regions. Therefore, in comparison to the case of the same stimulation and collection region, a hexagonal array would provide a decrease in total sweat rate of 70% (i.e., $100\% - (40\% Q_{\text{sub.0}} * 75\% A_{\text{sub.0}}) = 70\%$), resulting in a sweat rate of 30% of the original sweat rate, $Q_{\text{sub.0}}$, while using 25% of the original dose, $\Gamma_{\text{sub.0}}$. This calculation completely excludes any sweat generated underneath the stimulation region. Overall however, the SE ratio would increase 20% as a result of the reduced stimulation area (i.e., $0.30 Q_{\text{sub.0}} / 0.25 \Gamma_{\text{sub.0}} = 1.2 SE_{\text{sub.0}}$). Further, if one is able to collect the sweat generated underneath the stimulation region, the SE could increase 120% (i.e., $(0.30 Q_{\text{sub.0}} + 0.25 Q_{\text{sub.0}}) / 0.25 \Gamma_{\text{sub.0}} = 2.2 SE_{\text{sub.0}}$). Thus, not only is the contamination reduced with this method, but the sampling efficiency (SE) is increased. It should be noted that this reduction in area and ‘dose’ applies to all methods of stimulation where these nerve fibers are activated. Therefore, thermal, direct-electrical or other methods would also benefit from this unique device structure.

(31) In alternative embodiments, the stimulation region could be much smaller than the collection/sensor region. This is because only a small area would be required to initiate the SAR response of many nearby sweat glands. Furthermore, in the case of iontophoresis, this reduction in the stimulation area would improve the SE, which would allow for less drug delivery (dosage) to achieve the same previously attained SE.

(32) Dead Volume

(33) A metric for comparing the dead volume (“DV”) of such devices can be defined by Equation 2:

$$(34) DV = \frac{V_{\text{dead}}}{A_{\text{sensor}}}$$

where $V_{\text{sub.dead}}$ is the dead volume between the sensor and skin, and $A_{\text{sub.sensor}}$ is the area of the sensor. Comparing to an instance where the stimulation and collection regions are in the same area and the device includes an element to reduce contamination (not shown), such as a membrane, the gap between the sensor and skin could be on the order of 200 μm (e.g., due to roughness of skin, thickness of paper wicking layer, thickness of a drug reservoir, etc.). However, with indirect stimulation, one can greatly improve the contact between the skin and sensor while reducing the contamination from the stimulation. In this regard, the device may be so intimate with the skin that the only source of dead volume would be from the topology of the skin (e.g., about 30 μm). Thus, the reduction in dead volume between the device where the stimulation and collection regions are in the same area and this embodiment would be a reduction of at least $6\times$ ($200 \mu\text{m} / 30 \mu\text{m}$). This holds great significance when one estimates the time to “refresh” the sweat underneath the sensor. If there is $6\times$ less volume to refill underneath the sensor, then for a given flow rate, the time required to refresh the sweat underneath the sensor would also be reduced $6\times$. This has profound impact on time-resolution capabilities.

Total Figure of Merit (TOT-FOM)

(35) Utilizing the two calculations above for SE and DV and assuming the addition of an isolating membrane between sensor and skin, the effective improvement may be the multiplication of these two values. Therefore, the total improvement between an exemplary previous device described above and an embodiment of the present invention would be an improvement of at least $7.2\times$ with potentially an improvement approaching $13.2\times$ —when sweat rate per unit area, dose per unit area and dead volume per unit area are considered.

(36) While all of the invention has been illustrated by a description of various embodiments and while these embodiments have been described in considerable detail, it is not the intention of the Applicants to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications will readily appear to those skilled in the art. The invention in its broader aspects is therefore not limited to the specific details, representative apparatus and method, and

illustrative examples shown and described. Accordingly, departures may be made from such details without departing from the spirit or scope of the Applicants' general inventive concept.

Claims

1. A wearable sweat sensing device, comprising: a sweat stimulator configured to be placed on a surface of a first region of a device wearer's skin to directly stimulate sweating in the first region and to indirectly stimulate sweating in a second region of the wearer's skin, wherein the second region is distinct from the first region; a sweat analyzer configured to be placed on a surface of the second region of the device wearer's skin, the sweat analyzer configured to collect a sweat sample from the second region, and to measure one or more analytes in the sweat sample; and an isolation layer positioned to fluidically isolate the sweat stimulator from the sweat analyzer.
 2. The device of claim 1, wherein the sweat stimulator is configured to have a minimum area in contact with the device wearer's skin, a first dimension, and a second dimension, and wherein the ratio of the first dimension to the second dimension is one or more of the following: greater than 1 to 1, 2 to 1, or greater than 2 to 1.
 3. The device of claim 1, wherein the sweat stimulator is configured to have a surface area in contact with the device wearer's skin, and a minimum volume relative to the surface area.
 4. The device of claim 1, wherein the sweat stimulator is configured to use a pair of iontophoresis electrodes to drive a sweat stimulant into the first region.
 5. The device of claim 4, wherein a return electrode is located on a periphery of the device.
 6. The device of claim 4, wherein the sweat stimulant comprises a chemical stimulant selected from a group consisting of acetylcholine, methacholine, nicotine, and carbachol.
 7. The device of claim 1, wherein the sweat stimulator is electrically isolated from the sweat analyzer.
 8. The device of claim 1, wherein the sweat analyzer further comprises a collector configured to collect and convey the sweat sample from the second region to the sweat analyzer.
 9. The device of claim 8, wherein the collector further comprises a wicking material.
 10. The device of claim 1, wherein the first region is separated from the second region by a distance of 0.1 mm to 30 mm.
 11. The device of claim 1, further comprising a plurality of sweat stimulators and one or more sweat analyzers, and wherein the plurality of sweat stimulators and the one or more sweat analyzers are spatially arranged to increase the probability of a SAR sweating response underneath the one or more sweat analyzers.
 12. The device of claim 1, further comprising a plurality of sweat stimulators and a plurality of sweat analyzers, and wherein the plurality of sweat stimulators and the plurality of sweat analyzers are arranged spatially in a hexagonal pattern.
 13. The device of claim 11, wherein the plurality of sweat stimulators and the one or more sweat analyzers are arranged in a concentric pattern.
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